Result	Size	Time	Cycles	GPU	SM Frequency	Process	Attributes
Baseline 1 — 572 - convolution3D_kernel	(24, 24, 1)x(32, 32, 1)	45.06 us	72,913	0 - NVIDIA GeForce RTX 4060 Laptop GPU	1.62 Ghz	[104238] conv3d_baseline	

Summary | Details | Source | Context | Comments | Raw | Session — Compare | Tools | View | Export

GPU Speed Of Light Throughput

GPU Throughput Chart

High-level overview of the throughput for compute and memory resources of the GPU. For each unit, the throughput reports the achieved percentage of utilization with respect to the theoretical maximum. Breakdowns show the throughput for each individual sub-metric of Compute and Memory to clearly identify the highest contributor. High-level overview of the utilization for compute and memory resources of the GPU presented as a roofline chart.

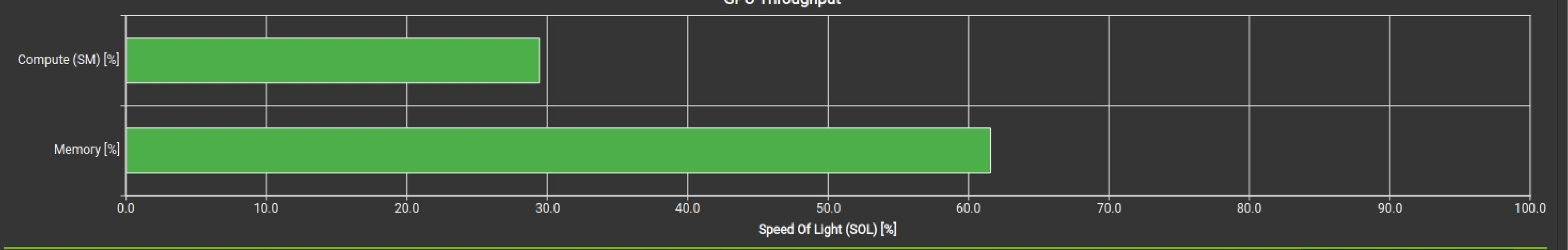
Compute (SM) Throughput [%]	29.43	Duration [us]	45.06
Memory Throughput [%]	61.57	Elapsed Cycles [cycle]	72,913
L1/TEX Cache Throughput [%]	32.47	SM Active Cycles [cycle]	66,083.33
L2 Cache Throughput [%]	19.00	SM Frequency [Ghz]	1.62
DRAM Throughput [%]	61.57	DRAM Frequency [Ghz]	7.98

High Memory Throughput Memory is more heavily utilized than Compute: Look at the Memory Workload Analysis section to identify the DRAM bottleneck. Check memory replay (coalescing) metrics to make sure you're efficiently utilizing the bytes transferred. Also consider whether it is possible to do more work per memory access (kernel fusion) or whether there are values you can (re)compute.

Key Performance Indicators

Roofline Analysis The ratio of peak float (FP32) to double (FP64) performance on this device is 64:1. The workload achieved 4% of this device's FP32 peak performance and 0% of its FP64 peak performance. See the Profiling Guide for more details on roofline analysis.

GPU Throughput



Compute Workload Analysis

Pipe Utilization (Elapsed Cycles)

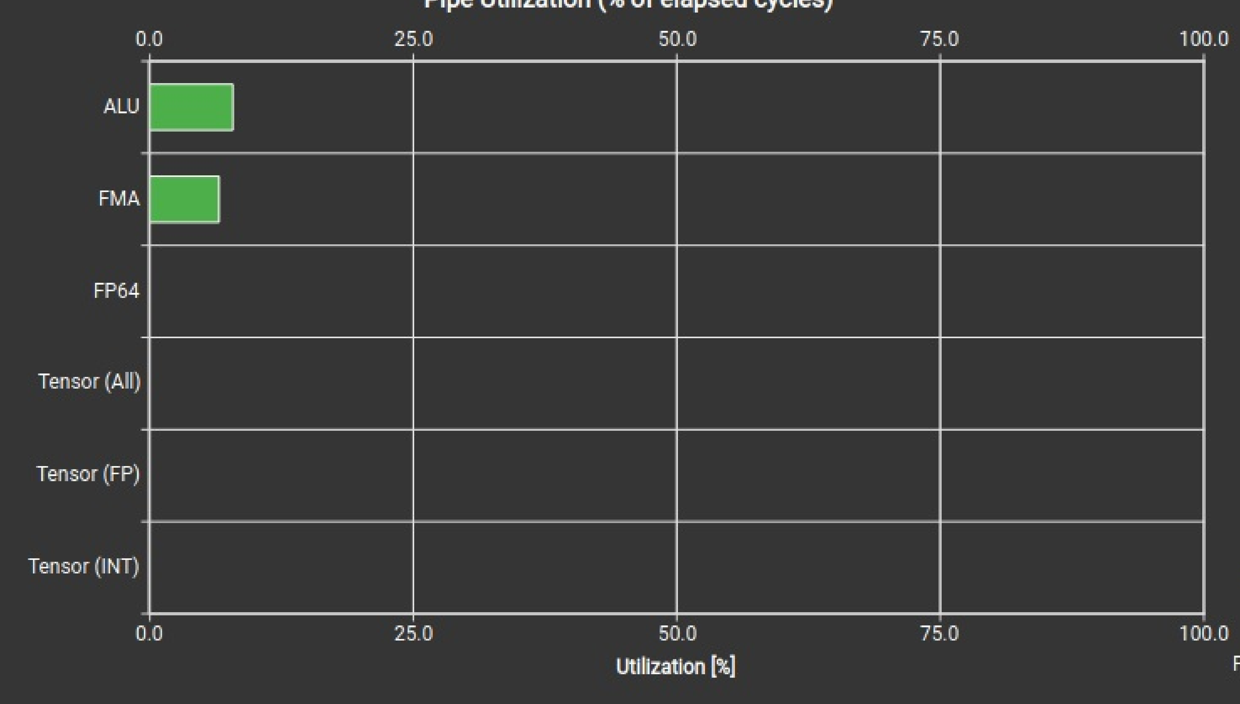
Detailed analysis of the compute resources of the streaming multiprocessors (SM), including the achieved instructions per clock (IPC) and the utilization of each available pipeline. Pipelines with very high utilization might limit the overall performance.

Executed Ipc Elapsed [inst/cycle]	0.59	SM Busy [%]	14.83
Executed Ipc Active [inst/cycle]	0.65	Issue Slots Busy [%]	14.83
Issued Ipc Active [inst/cycle]	0.65		

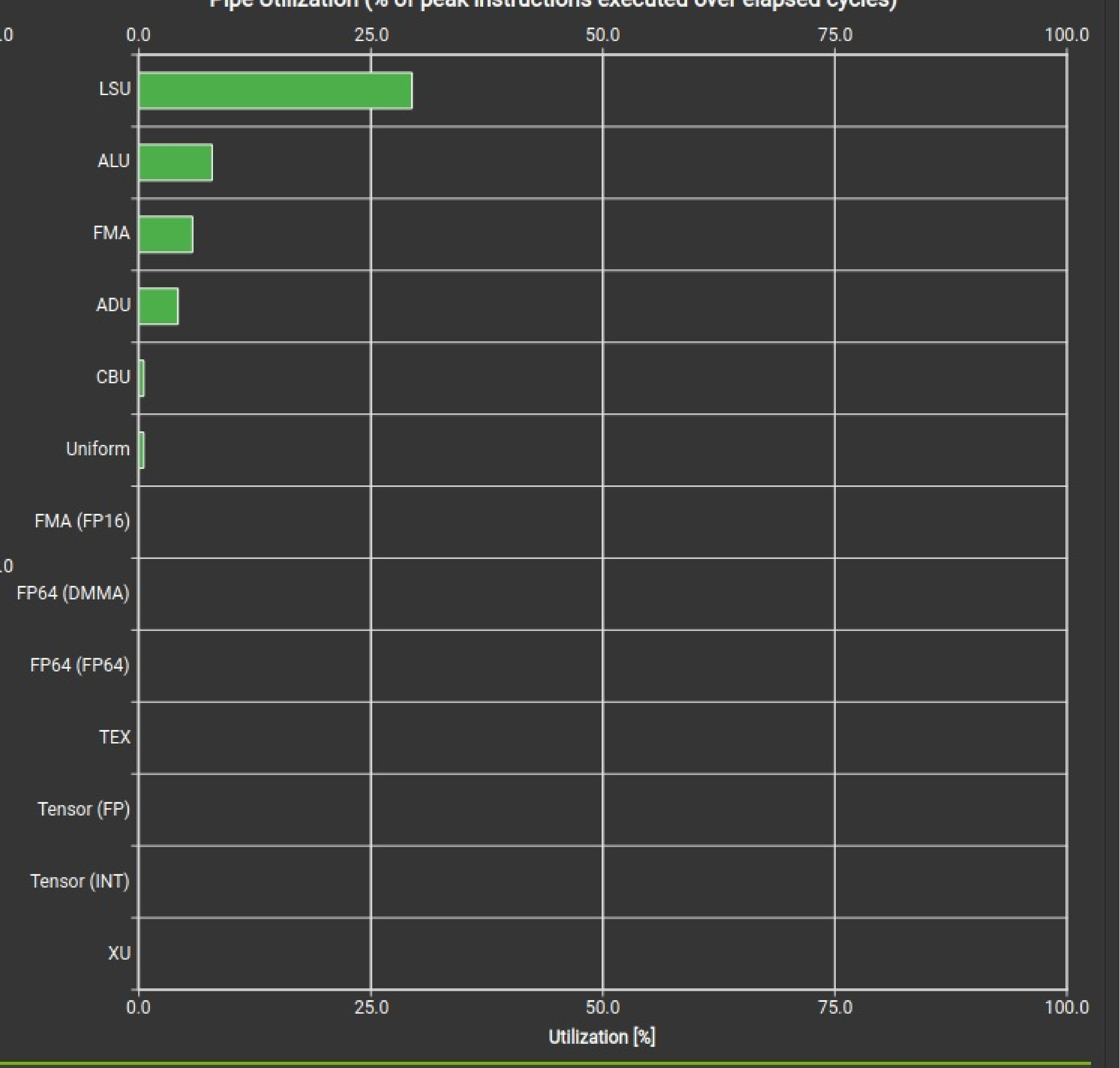
Low Utilization
Est. Local Speedup: 92.10% All compute pipelines are under-utilized. Either this workload is very small or it doesn't issue enough warps per scheduler. Check the Launch Statistics and Scheduler Statistics sections for further details.

Key Performance Indicators

Pipe Utilization (% of elapsed cycles)



Pipe Utilization (% of peak instructions executed over elapsed cycles)



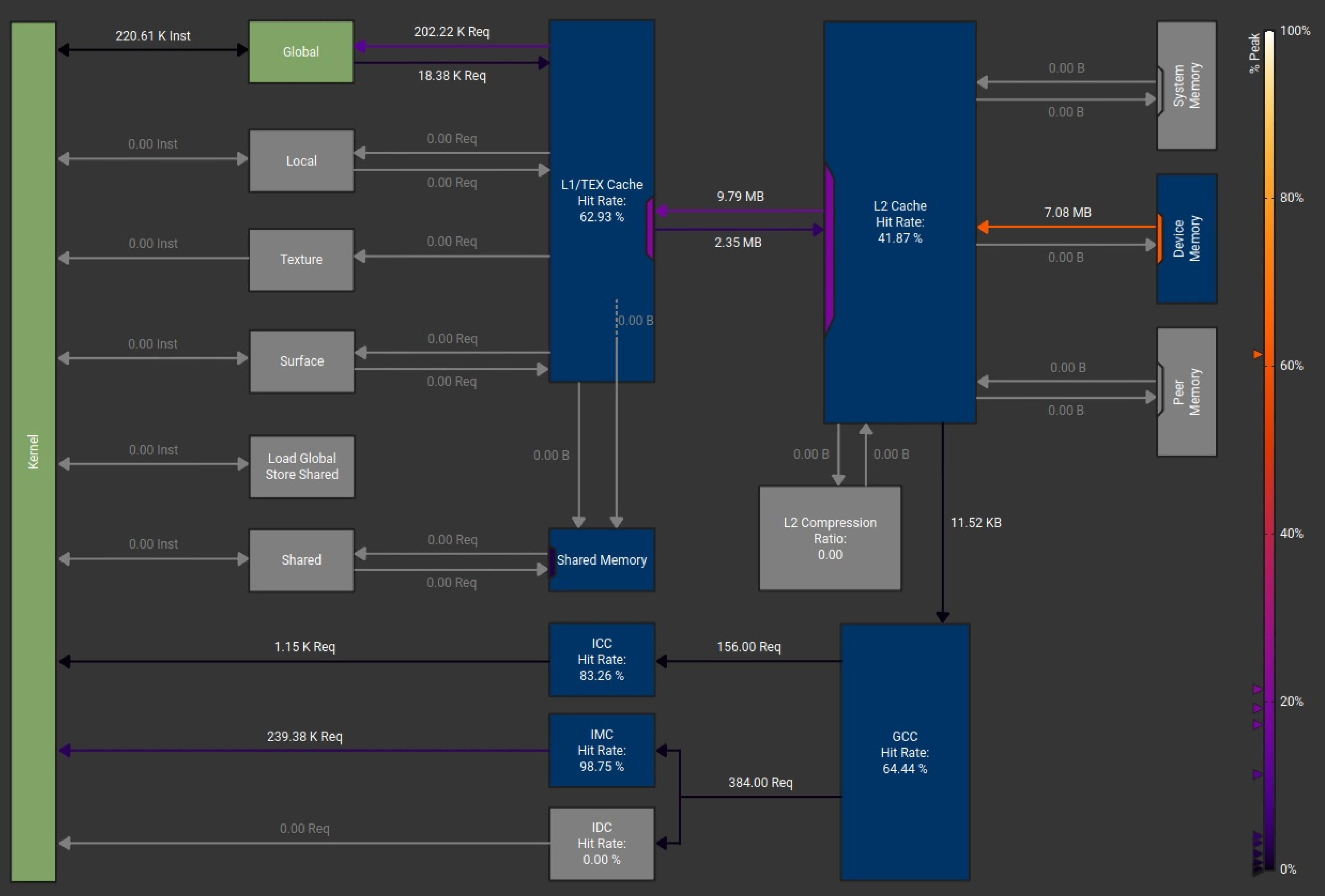
Memory Workload Analysis

Detailed analysis of the memory resources of the GPU. Memory can become a limiting factor for the overall kernel performance when fully utilizing the involved hardware units (Mem Busy), exhausting the available communication bandwidth between those units (Max Bandwidth), or by reaching the maximum throughput of issuing memory instructions (Mem Pipes Busy). Detailed chart of the memory units.

Memory Throughput [Gbyte/s]	157.17	Mem Busy [%]	16.37
L1/TEX Hit Rate [%]	62.93	Max Bandwidth [%]	61.57
L2 Hit Rate [%]	41.87	Mem Pipes Busy [%]	29.43
L2 Compression Input Sectors [sector]	0	Local Memory Spilling Requests	0
L2 Compression Ratio	0	Local Memory Spilling Request Overhead [%]	0
L2 Compression Success Rate [%]	0	L2 Persisting Size [Mbyte]	6.29

Memory Chart

Values: Transfer Size Inactivity: Greyed Out



Launch Statistics

Summary of the configuration used to launch the kernel. The launch configuration defines the size of the kernel grid, the division of the grid into blocks, and the GPU resources needed to execute the kernel. Choosing an efficient launch configuration maximizes device utilization.

Grid Size	576	Function Cache Configuration	CachePreferNone
Registers Per Thread [register/thread]	32	Static Shared Memory Per Block [byte/block]	0
Block Size	1,024	Dynamic Shared Memory Per Block [byte/block]	0
Threads [thread]	589,824	Driver Shared Memory Per Block [Kbyte/block]	1.02
Waves Per SM	24	Shared Memory Configuration Size [Kbyte]	8.19
Uses Green Context	0	Stack Size	1,024
# SMs [SM]	24	# TPCs	12
Enabled TPC IDs	all	-	-

Occupancy

% Occupancy Graphs

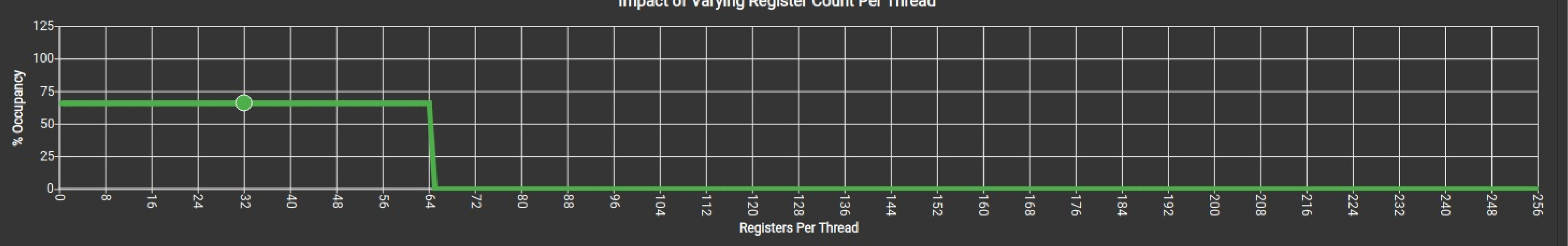
Occupancy is the ratio of the number of active warps per multiprocessor to the maximum number of possible active warps. Another way to view occupancy is the percentage of the hardware's ability to process warps that is actively in use. Higher occupancy does not always result in higher performance, however, low occupancy always reduces the ability to hide latencies, resulting in overall performance degradation. Large discrepancies between the theoretical and the achieved occupancy during execution typically indicates highly imbalanced workloads.

Theoretical Occupancy [%]	66.67	Block Limit Registers [block]	2
Theoretical Active Warps per SM [warp]	32	Block Limit Shared Mem [block]	8
Achieved Occupancy [%]	57.17	Block Limit Warps [block]	1
Achieved Active Warps Per SM [warp]	27.44	Block Limit SM [block]	24

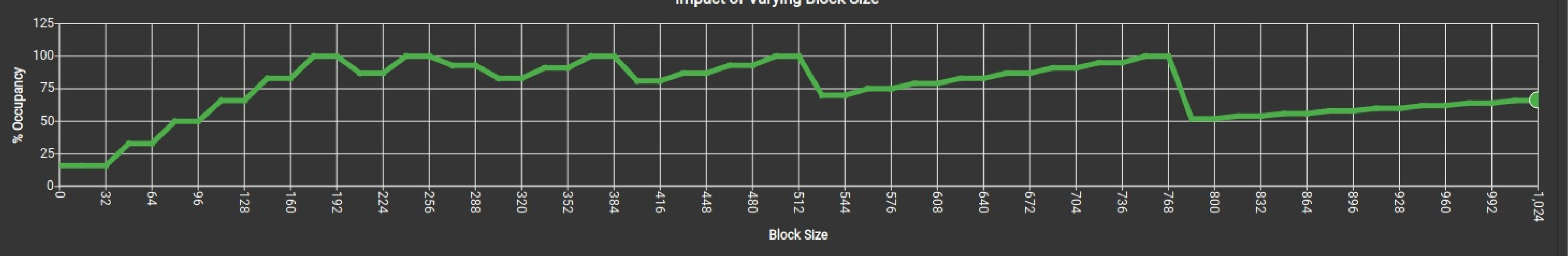
Theoretical Occupancy
Est. Local Speedup: 33.33% The 8.00 theoretical warps per scheduler this kernel can issue according to its occupancy are below the hardware maximum of 12. This kernel's theoretical occupancy (66.7%) is limited by the number of warps within each block.

Key Performance Indicators

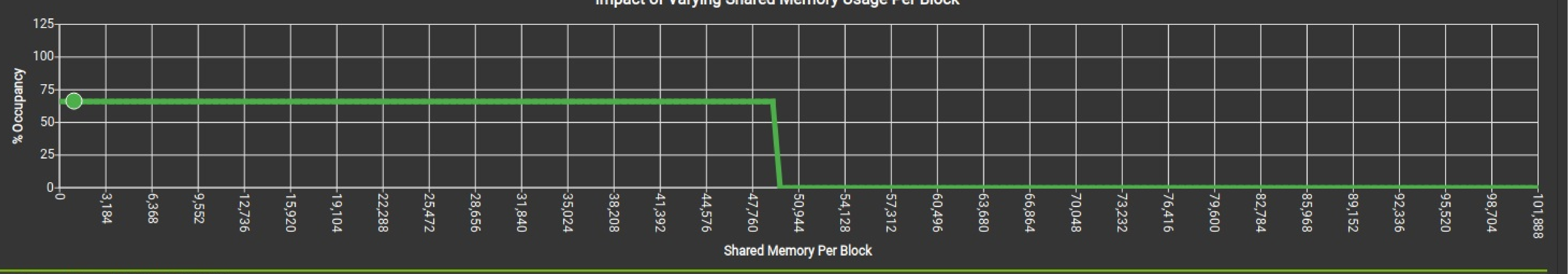
Impact of Varying Register Count Per Thread



Impact of Varying Block Size



Impact of Varying Shared Memory Usage Per Block



GPU and Memory Workload Distribution

Analysis of workload distribution in active cycles of SM, SMP, SMSP, L1 & L2 caches, and DRAM

Average SM Active Cycles [cycle]	66,083.33	Average L1 Active Cycles [cycle]	66,083.33
Average L2 Active Cycles [cycle]	61,423.50	Average SMSP Active Cycles [cycle]	63,583.22
Average DRAM Active Cycles [cycle]	221,300	Total SM Elapsed Cycles [cycle]	1,749,736
Total L1 Elapsed Cycles [cycle]	1,749,736	Total L2 Elapsed Cycles [cycle]	1,164,992
Total SMSP Elapsed Cycles [cycle]	6,998,944	Total DRAM Elapsed Cycles [cycle]	1,437,696

Workload Distribution

	Average	Min	Max	Sum
SM Active Cycles	66,083.33	65,071	66,725	1,586,000
SMSP Active Cycles	63,583.22	61,450	65,174	6,103,989
L1 Active Cycles	66,083.33	65,071	66,725	1,586,000
L2 Active Cycles	61,423.50	60,431	62,187	982,776
DRAM Active Cycles	221,300	221,216	221,344	885,200

Source Counters

Source metrics, including branch efficiency and sampled warp stall reasons. Warp Stall Sampling metrics are periodically sampled over the kernel runtime. They indicate when warps were stalled and couldn't be scheduled. See the documentation for a description of all stall reasons. Only focus on stalls if the schedulers fail to issue every cycle.

Branch Instructions [inst]	36,816	Branch Efficiency [%]	0
Branch Instructions Ratio [%]	0.04	Avg. Divergent Branches [branches]	0

Uncoalesced Global Accesses
Est. Speedup: 11.62% This kernel has uncoalesced global accesses resulting in a total of 140944 excessive sectors (14% of the total 1023376 sectors). Check the L2 Theoretical Sectors Global Excessive table for the primary source locations. The CUDA Programming Guide has additional information on reducing uncoalesced device memory accesses.

Key Performance Indicators

L2 Theoretical Sectors Global Excessive

Location	Value	Value (%)
0x7572f3272550 in convolution3D_kernel	17,618	13
0x7572f3272540 in convolution3D_kernel	17,618	13
0x7572f3272530 in convolution3D_kernel	17,618	13
0x7572f3272520 in convolution3D_kernel	17,618	13
0x7572f3272510 in convolution3D_kernel	17,618	13

Warp Stall Sampling (All Samples)

Location	Value	Value (%)
0x7572f3272570 in convolution3D_ke...	310	20
0x7572f32725d0 in convolution3D_ke...	290	19
0x7572f3272580 in convolution3D_ke...	152	10
0x7572f32725e0 in convolution3D_ke...	138	9
0x7572f3272680 in convolution3D_ke...	73	5

Most Instructions Executed

Location	Value	Value (%)
0x7572f3272420 in convolution3D_ke...	18,432	2
0x7572f3272410 in convolution3D_ke...	18,432	2
0x7572f3272400 in convolution3D_ke...	18,432	2
0x7572f32723f0 in convolution3D_ker...	18,432	2
0x7572f32723e0 in convolution3D_ke...	18,432	2

Still missing data? Collect the *full* metric set, enable individual sections or learn how to select individual metrics for collection.